OCR Further Mathematics
Pure Core - Hyperbolic Functions
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$y = 4\sinh x + 3\cosh x$ $\Rightarrow \frac{dy}{dx} = 4\cosh x + 3\sinh x$ $= 0$ when $4\cosh x + 3\sinh x = 0$ $\Rightarrow 4\left(\frac{e^x + e^{-x}}{2}\right) + 3\left(\frac{e^x - e^{-x}}{2}\right) = 0$ $\Rightarrow e^{2x} = -\frac{1}{7}$ which is not possible as $e^{2x} > 0$ so no turning points Alternative method $y = 4\sinh x + 3\cosh x$ $\Rightarrow \frac{dy}{dx} = 4\cosh x + 3\sinh x$ $= 0$ when $\tanh x = -\frac{4}{3}$ But $\lvert\tanh x\rvert < 1$ for all x. So there are no values of x for which $\frac{dy}{dx} = 0$ So no turning points	M1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.4) Alternative method M1 M1 A1 [3]	Diffn (Hyperbolics or exponentials) Set = 0 and use exponential forms – can change to exponentials before diffn. Conclusion with justification Alternative method Differentiate Set = 0 and use formula for tanh Conclusion with justification	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$y = 4\sinh x + 3\cosh x = 5$ $\Rightarrow 4\left(\frac{e^x - e^{-x}}{2}\right) + 3\left(\frac{e^x + e^{-x}}{2}\right) = 5$ $\Rightarrow 7e^x - e^{-x} = 10 \Rightarrow 7e^{2x} - 10e^x - 1 = 0$ $e^x = \frac{10 \pm \sqrt{100+28}}{14} = \frac{5+\sqrt{32}}{7}$ or $\frac{5-\sqrt{32}}{7}$ But $e^x > 0$ so cannot $= \frac{5-\sqrt{32}}{7}$ So the only root is $e^x = \frac{5+\sqrt{32}}{7}$ $\Rightarrow x = \ln\left(\frac{5+4\sqrt{2}}{7}\right)$ Alternative method (see appendix for full working) $4\sinh x + 3\cosh x = 5 \Rightarrow 4\sinh x = 5 - 3\cosh x$ $\therefore 16\sinh^2 x = 16(\cosh^2 x - 1) = 25 - 30\cosh x + 9\cosh^2 x$ $7\cosh^2 x + 30\cosh x - 41 = 0$ $\cosh x \geq 1 \Rightarrow$ cosh	M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 2.3) A1 (AO 1.1) Alternative method M1 M1 A1 A1 A1 [5]	Use of exponentials equation of the form $ae2^x + be^x + c$ (for non-zero a, b and c) Two roots for e^x One rejected plus reason Ignore inclusion of 2nd root Alternative method Use Pythagoras Quadratic in cosh (or sinh) Two roots One rejected plus reason	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$x = \dfrac{-15+16\sqrt{2}}{7}$ $\Rightarrow x = \cosh^{-1}\dfrac{-15+16\sqrt{2}}{7} = \pm\ln\left(\dfrac{-15+16\sqrt{2}+4\sqrt{43-30\sqrt{2}}}{7}\right)$ But the negative root does not work in the original equation since LHS would be negative while RHS would be positive (but equal when squared). $\therefore x = \ln\left(\dfrac{-15+16\sqrt{2}+4\sqrt{43-30\sqrt{2}}}{7}\right)$			
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2	a		DR $\frac{x^3+x^2+9x-1}{x^3+x^2+4x+4} \equiv \frac{x^3+x^2+4x+4+5x-5}{x^3+x^2+4x+4}$ $=1+\frac{5x-5}{x^3+x^2+4x+4}$ So A = 1, B = 5 and C = −5	B1 (AO3.1a) [1]	Attempt to divide out improper fraction. Could be by symbolic division or other valid method (eg comparing coefficients or substitution of values for x)	Allow embedded answers
	b		DR $x^3+x^2+4x+4=(x+1)(x^2+4)$ $\frac{5x-5}{x^3+x^2+4x+4}=\frac{D}{x+1}+\frac{Ex+F}{x^2+4}$ $D(x^2+4)+(x+1)(Ex+F)=5x-5$ $x = - \Rightarrow 5D = -10 \Rightarrow D = -2$ $x = 0 \Rightarrow -2-F = -5 \Rightarrow F = -3$ x^2: $D + E = 0 \Rightarrow E = 2$ $1-\frac{2}{x+1}+\frac{2x+3}{x^2+4}$	B1 (AO3.1a) M1 (AO1.2) A1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [5]	Correct factorisation of cubic seen in working Correct form for partial fractions equated to their remainder rational fraction from (a). Follow through their division and factorisation. Or equivalent to find D correctly.	Could be from improper fraction Allow ft A1 for second and third coefficients found. Or $1-\frac{2}{x+1}+\frac{2x}{x^2+4}+\frac{3}{x^2+4}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		DR $\int_0^2 \frac{x^3+x^2+9x-1}{x^3+x^2+4x+4}\,dx = \int_0^2 1-\frac{2}{x+1}+\frac{2x}{x^2+4}+\frac{3}{x^2+4}\,dx$ $=\left[x-2\ln(x+1)+\ln(x^2+4)+\frac{3}{2}\tan^{-1}\left(\frac{x}{2}\right)\right]_0^2$ $\left(2-2\ln 3+\ln 8+\frac{3\pi}{8}\right)-\ln 4$ $2+\ln\left(\frac{2}{9}\right)+\frac{3}{8}\pi$	*M1 (AO3.1a) dep*M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	Split term with $x^2 + 4$ in denominator and $ax + b$ in numerator Correctly integrate *their* expression (ignore limits) Correctly substitute limits to produce exact values and evaluate their $\tan^{-1}$ term $a = 2$, $b = \frac{2}{9}$, $c = \frac{3}{8}$	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a		DR $\text{RHS} = 2\cosh^2 x - 1 = 2\left(\frac{e^x + e^{-x}}{2}\right)^2 - 1$ $= 2\left(\frac{e^{2x} + 2 + e^{-2x}}{4}\right) - 1 = \frac{e^{2x} + 2 + e^{-2x}}{2}$ $= \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x = \text{LHS}$	M1 (AO2.1) A1 (AO2.1) [2]	Uses correct exponential form in an attempt at proof AG	Proof must be complete	
	b		DR $2\cosh^2 x - 1 = 3\cosh x + 1$ $\Rightarrow 2\cosh^2 x - 3\cosh x - 2 = 0$ $(2\cosh x + 1)(\cosh x - 2) = 0$ $\cosh x = 2$ or $-½$ $\cosh x \geq 1$ so $\neq -½$ $x = \cosh^{-1} 2 = \ln\left(2 + \sqrt{3}\right)$ $x = \ln\left(2 - \sqrt{3}\right)$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO2.3) A1 (AO1.1) A1 (AO1.1) [6]	Use of identity in above part to leave a three term quadratic equation in just cosh$x$ Attempt to solve eg $\frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 2 \times (-2)}}{2 \times 2}$ Justification must be seen and must contain no incorrect statements For either correct answer seen Both correct values for x	or $2\left(\cosh x - \frac{3}{4}\right)^2 - \frac{9}{8} - 2 = 0$ Or solves quadratic BC $x = -\ln\left(2 + \sqrt{3}\right)$ Or Mark final answer	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	8	

Question			Answer/Indicative content	Marks	Part marks and guidance		
4			DR $\frac{1}{2}\int\left(\sqrt{sin\theta}e^{\frac{1}{3}cos\theta}\right)^2 \mathrm{d}\theta$ $A=\frac{1}{2}\int_0^{\pi}\sin\theta e^{\frac{2}{3}\cos\theta}\mathrm{d}\theta$ $=\frac{1}{2}\times-\frac{3}{2}\left[e^{\frac{2}{3}\cos\theta}\right]_0^{\pi}$ $\frac{3}{4}\left(e^{\frac{2}{3}}-e^{-\frac{2}{3}}\right)$	M1(AO3.1a) *A1(AO2.1) dep*M1(AO1.1a) A1(AO1.1)[4]	Correct form, in terms of θ, Integrand has been squared out. Must include limits (can be seen later) Might be as result of substitution Allow coefficient error for M1 isw	M1 can be implied by 1.0757… BC eg $\frac{3}{4}\left[e^{u}\right]_{-\frac{2}{3}}^{\frac{2}{3}}$ or $\frac{3}{4}\left[e^{\frac{2}{3}u}\right]_{-1}^{1}$ oe	
					Examiner's Comments Where an integrand is the product of two functions, there are three techniques available to candidates - inspection, substitution and parts. Candidates at this level should be encouraged to develop their 'inspection' skills sufficiently to recognise this integrand to within a constant term. The purpose of the other two techniques is to produce an integrand that has become simpler to integrate. Where candidates selected 'parts' (which a lot did) they produced a more complicated integrand; at this point they should reconsider their approach. Flexibility of approach to problem-solving is an important skill to develop.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	4		
5			$\int 1\times\sin^{-1}x\,dx = x\sin^{-1}x - \int\frac{x}{\sqrt{1-x^2}}dx$ $= x\sin^{-1}x + \left(1-x^2\right)^{\frac{1}{2}}$ (+c) $\int_0^{\frac{1}{2}} f(x) = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) [3]	Attempt integration by parts <u>Examiner's Comments</u> In (a)(ii) the command word "determine" notifies candidates that they cannot merely use their calculator to find f'''(0). Of course, the calculator could be used here (and in questions such as Question 1) to double check responses and alert candidates as to errors they may have made. In (b) while most candidates were able to gain the first mark by attempting integration by parts, several were unable to tackle the integral of $\frac{x}{\left(1-x^2\right)^{\frac{1}{2}}}$. Centres may find it helpful to allow plenty of practice of selecting a suitable method of integration when facing such expressions. Candidates should then be alert to the significance, here, of 'x' being the derivative of 'x^2' (to within a constant term).	ignore limits. Formula for parts must be correct
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
6	a		Min value of cosh is 1 (and point on ground is at the minimum) (so $0 = k \times 1 - 1 \Rightarrow$) $k = 1$	M1 (AO2.2a) A1 (AO2.2a) [2]	Using minimum point of curve and knowledge of cosh graph	Could be derived by differentiation If zero scored then sc1 for k=1 www	
	b		Passes through $(0, 3) \Rightarrow 3 = \cosh(-b) - 1$	*M1 (AO3.3)	Use of (0, 3) to derive an expression for b	accept $\cosh(-b) = \frac{4}{k}$	
			$\Rightarrow b = -\cosh^{-1}(3 + 1)$ $b = (\pm)\ln(4 + \sqrt{(4^2 - 1)})$	dep*M1 (AO3.1a)	Correct numerical use of formula	Or rearranges.	
			$\Rightarrow b = \ln(4 + \sqrt{15})$	A1 (AO1.1)			
			Passes through $(2, 0) \Rightarrow 0 = \cosh(2a - b) - 1$ $\Rightarrow b = 2a$	MK1 (AO3.3)	Use of (2, 0) to derive $b = 2a$	Could be from (a). Allow ft	
			$\Rightarrow a = ½\ln(4 + \sqrt{15})$	A1 (AO1.1) [5]			
	c		(By symmetry of both;) (4, 3)	B1 (AO2.2a) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		Holly's model; $d_H = 6.75$	B1 (AO3.4)		Condone 27/4	
			Jofra's model: $d_J = \cosh(5a - b) - 1$	M1 (AO3.4)	Use of $x = 5$ with their values of a and b to predict d. Must have $- 1$	$a = 1.0317\ldots$, $b = 2.0634\ldots$, $d_J = 10.067\ldots$	
			AG $d_J - d_H = 10.067\ldots - 6.75 = 3.32$ (3 sf)	A1 (AO1.1) [3]	From correct values	Condone 10.07 only if clear evidence of production	
					Examiner's Comments Few candidates scored highly on this question which may well reflect the atypical cohort. It is often the case that mathematical models, at this level, are formed from the transformation of standard functions. Centres should give candidates plenty of practice in this approach so that they recognise the key features of the function in its transformed form. Here, the position of minimum value of cosh x is key.		
			Total	11			

Question			Answer/Indicative content	Marks	Part marks and guidance	
7	a		$2\sinh^2 u + 1 \equiv 2\left(\frac{e^u - e^{-u}}{2}\right)^2 + 1$	M1 (AO2.1)	Use of exponential form for sinh u	
			$= \frac{e^{2u} - 2 + e^{-2u}}{2} + 1 \equiv \frac{e^{2u} + e^{-2u}}{2} \equiv \cosh 2u$ AG	A1 (AO2.1) [2]		
	b		cosh $2u \equiv 2$ sinh^2 u + 1 $\Rightarrow$ 2 sinh $2u \equiv 4$ sinh u cosh u $\Rightarrow$ sinh $2u \equiv$ sinh u cosh u AG	B1 (AO2.1) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		$x = \sinh^2 u \Rightarrow dx = 2 \sinh u \cosh u \, du$	M1 (AO3.1a)	Attempt to find $\frac{dx}{du}$	
			$\Rightarrow \int \sqrt{\frac{x}{x+1}} dx = \int \sqrt{\frac{\sinh^2 u}{\sinh^2 u+1}} 2 \sinh u \cosh u \, du$			
			$= 2\int \sinh^2 u \, du$	A1 (AO1.1)		
			$= \int (\cosh 2u - 1) \, du$	M1 (AO1.1a)	Use double angle formulae and attempt to integrate.	
			$= \frac{1}{2} \sinh 2u - u + c = \sinh u \cosh u - u + c$	A1 (AO1.1)	Ignore c.	
			$= \sqrt{x(1+x)} - \sinh^{-1} \sqrt{x} + c$			
			So $f(x) = \sqrt{x(1+x)} + c$, $a = -1$, $b = 1$	A1 (AO1.1)	c must be included here as part of $f(x)$ – allow a and b not being stated explicitly but $f(x)$ must be	
			Alternative method			
			$= 2\int \left(\frac{e^u - e^{-u}}{2}\right)^2 du = \frac{1}{2}\int e^{2u} - 2 + e^{-2u} \, du$	M1 (AO1.1a)	Use exponentials and attempt to integrate.	
			$= \frac{1}{4}e^{2u} - \frac{1}{4}e^{-2u} - u + c = \frac{1}{2}\sinh 2u - u + c$	A1 (AO1.1)	Ignore c.	
			$= \sqrt{x(1+x)} - \sinh^{-1} \sqrt{x} + c$			
			So $f(x) = \sqrt{x(1+x)} + c$, $a = -1$, $b = 1$	A1 (AO1.1) [5]	c must be included here as part of $f(x)$ – allow a and b not being stated explicitly but $f(x)$ must be	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		Area $= \left[\sqrt{x(1+x)} - \sinh^{-1}\sqrt{x}\right]_1^2$ $=\left(\sqrt{6}-\ln(\sqrt{2}+\sqrt{3})\right)-\left(\sqrt{2}-\ln(1+\sqrt{2})\right)$ $=\sqrt{6}-\sqrt{2}+\ln\left(\dfrac{1+\sqrt{2}}{\sqrt{2}+\sqrt{3}}\right)$ So $p=\sqrt{6}-\sqrt{2}$, q = 1, $r=\dfrac{1+\sqrt{2}}{\sqrt{2}+\sqrt{3}}$	M1 (AO1.1) A1 (AO1.1) [2]	Correct limits substituted and subtracted into *their* answer to (c) soi *p*, *q*, *r* must be stated <u>Examiner's Comments</u> Part (a) was standard but a number of candidates found an identity in exponentials without mentioning or cosh 2*u*. or sinh*u*. In part (c) many did not do a complete substitution, ending with an integral in *u* to be integrated with respect to *x*. Those that progressed well here, recognising also that the result of part (a) was required, often did not complete the question to give the response in the form stated. Unless part (c) was correct, the response for part (d) would not take the form given, although a substitution into their response for part (c) gained the method mark.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
8	a		$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{2x}+1}{2e^x}$ $\Rightarrow \operatorname{sech} x = \frac{2e^x}{e^{2x}+1}$ AG	M1(AO1.1) A1(AO2.1) [2]	Use of cosh$x$ in exponentials Examiner's Comments This was done well by almost all candidates.		
	b		$u = e^x \Rightarrow du = e^x\,dx$ $\Rightarrow dx = \frac{du}{u}$ $\Rightarrow \int \operatorname{sech} x\,dx = \int \left(\frac{2e^x}{e^{2x}+1}\right)dx$ $= \int \frac{2u}{u^2+1}.\frac{du}{u}$ = 2 $\tan^{-1}(u)$ + c = 2 $\tan^{-1}(e^x)$ + c Alternatively: u = sinh$x$ ⇒ du = cosh$x$dx M1 $\Rightarrow \int \operatorname{sech} x dx = \int \frac{1}{\cosh x}.\frac{du}{\cosh x} = \int \frac{du}{\cosh^2 x}$ $= \int \frac{du}{1+\sinh^2 x} = = \int \frac{du}{1+u^2}$ A1 = $\tan^{-1} u + c$ M1 = $\tan^{-1}$ (sinh$x$) + c A1 Alternatively: $\int \operatorname{sech} x dx = \int \frac{2e^x}{e^{2x}+1}dx$ Let $e^x = \tan u \Rightarrow e^x dx =$ $\sec^2 u du \Rightarrow$ $dx = \frac{\sec^2 u}{\tan u}du$	M1(AO3.1a) A1(AO1.1) M1(AO3.1a) A1(AO1.1) [4]	Substitute and use (a) any form entirely in terms of u Use standard form for integral and substitute back Must include c	Allow absence of du	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			M1 $\Rightarrow \int \text{sech}x\,\text{d}x =$ $\int \frac{2\tan u}{\tan^2 u+1}\cdot\frac{\sec^2 u}{\tan u}\text{d}u = 2\int \text{d}u$ A1 $2u + c$ M1 $2\tan^{-1}(e^x) + c$ A1		Substitute In correct form Integrate and substitute back Must include c <u>Examiner's Comments</u> Most candidates made a good choice of substitution to obtain the indefinite integral. A number of candidates however lost the final accuracy mark by not including an arbitrary constant of integration.		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
9	a		$5\cosh x + 3\sinh x = 4$ $\Rightarrow 5\left(\frac{e^x + e^{-x}}{2}\right) + 3\left(\frac{e^x - e^{-x}}{2}\right) = 4$ $\Rightarrow 4e^x + e^{-x} = 4$ $\Rightarrow 4e^{2x} - 4e^x + 1 = 0$ $\left(\Rightarrow (2e^x - 1)^2 = 0\right)$ $\Rightarrow e^x = \frac{1}{2}$ $\Rightarrow x = -\ln 2$ oe Alternatively: $5\cosh x + 3\sinh x \equiv R\cosh(x + \alpha)$ M1 where $R = \sqrt{25-9} = 4$, $\tanh \alpha = \frac{3}{5} \Rightarrow \alpha = \tanh^{-1}\frac{3}{5} = \frac{1}{2}\ln\left(\frac{1+\frac{3}{5}}{1-\frac{3}{5}}\right) = \frac{1}{2}\ln 4 = \ln 2$ M1 A1 $\Rightarrow 4\cosh(x + \alpha) = 4 \Rightarrow \cosh(x + \alpha) = 0$ $\Rightarrow x = -\alpha = -\ln 2$ A1	M1(AO3.1a) M1(AO3.1a) A1(AO1.1) A1(AO1.1) [4]	Use of exponentials Multiply by e^x **<u>Examiner's Comments</u>** This question was answered well; only a handful of candidates wrote '$\cosh x = e^x + e^{-x}$' instead of '$\cosh x = \frac{1}{2}(e^x + e^{-x})$' and similarly for $\sinh x$. There were other approaches to this question, including the compound angle formulae and the method of squaring and using Pythagoras' theorem; these were usually less successful.	Alternatively make cosh the subject, square and use Pythagoras to give quadratic in cosh Alternatively use compound angle formula	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		DR $\int_{-1}^{1}(5\cosh x+3\sinh x)\,dx=[5\sinh x+3\cosh x]_{-1}^{1}$ $=\left(5\frac{e^1-e^{-1}}{2}+3\frac{e^1+e^{-1}}{2}\right)-\left(5\frac{e^{-1}-e^1}{2}+3\frac{e^{-1}+e^1}{2}\right)$ $=(4e^1-e^{-1})-(4e^{-1}-e^1)$ $=5e-\frac{5}{e}$ Alternatively: 5cosh$x$ + 3 sinh$x$ $=5\left(\frac{e^x+e^{-x}}{2}\right)+3\left(\frac{e^x-e^{-x}}{2}\right)=\frac{1}{2}(8e^x+2e^{-x})$ M1 $\Rightarrow\int_{-1}^{1}\frac{1}{2}(8e^x+2e^{-x})\,dx=\left[4e^x-e^{-x}\right]_{-1}^{1}$ M1 $=(4e-e^{-1})-(4e^{-1}-e)=5e-\frac{5}{e}$ A1	M1(AO1.1) M1(AO1.1) A1(AO2.1) [3]	Attempt at integral (i.e. one function changed) Convert sinh$x$ and cosh$x$ to exponential form in *their* integrated function and use limits correctly <u>Examiner's Comments</u> The conversion to exponential form could be done before or after integration. A few candidates made sign errors. It was surprising to see so many candidates choosing to state that a = 5 and b = 5, having obtained the right answer.	Alternatively: M1 convert (including possibly using result from (a)) M1 integrate and use limits correctly
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		a	$\frac{dx}{dt} = y \Rightarrow \frac{d^2x}{dt^2} = \frac{dy}{dt}$	M1 (AO 3.4)	Differentiate 1st DE and substitute back in	
			$= 2y - 5x$			
			$\Rightarrow \frac{d^2x}{dt^2} = 2\frac{dx}{dt} - 5x$	A1 (AO 2.1)	AG	
				[2]		
		b	(i) $\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + 5x = 0$			
			A.E. is $n^2 - 2n + 5 = 0$	M1 (AO 1.1)	For auxiliary equation	
			$\Rightarrow n = 1 \pm 2i$	A1 (AO 1.1)	BC	
			$\Rightarrow x = e^t (A \sin 2t + B \cos 2t)$	A1 (AO 2.2a)		
				[3]		
			(ii) $x = e^t (A \sin 2t + B \cos 2t)$ and $\frac{dx}{dt} = y$	M1 (AO 1.1)		
			$\Rightarrow \frac{dx}{dt} = e^t (A \sin 2t + B \cos 2t) + e^t (2A \cos 2t - 2B \sin 2t)$			
			$- y = e^t ((A - 2B) \sin 2t + (2A + B) \cos 2t)$	A1 (AO 2.2a)		
				[2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$x = e^t (A \sin 2t + B \cos 2t)$ $x = 1, t = 0$ $\Rightarrow 1 = B$ $\Rightarrow y = e^t ((A - 2)\sin 2t + (2A + 1) \cos 2t)$ $y = 3, t = 0$ $\Rightarrow 3 = 2A + 1 \Rightarrow A = 1$ $\Rightarrow x = e^t (\sin 2t + \cos 2t), y = e^t (3 \cos 2t - \sin 2t)$	M1 (AO 3.3) A1 (AO 1.1) M1 (AO 3.3) A1 (AO 1.1) A1 (AO 3.3) [5]	Translate context into condition to find B Translate context into condition to find A Both stated with A and B substituted for	
		d	$x = 0 \Rightarrow x = e^t (\sin 2t + \cos 2t) = 0$ $t = 1.178...$ So 1.12 year $y = 0 \Rightarrow y = e^t (3 \cos 2t - \sin 2t) = 0$ $t = 0.624...$ So 0.625 year 0.625 < 1.12 so the prey die out first.	M1 (AO 3.1b) A1 (AO 3.4) A1 (AO 3.4) E1 (AO 3.4) [4]	Set either their $x = 0$ or their $y = 0$ and solve BC BC	
			Total	16		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		a	$\tanh x = \dfrac{e^x - e^{-x}}{e^x + e^{-x}} = u$ $\Rightarrow e^x(1 - u) = e^{-x}(1 + u)$ $\Rightarrow e^{2x} = \left(\dfrac{1+u}{1-u}\right) \Rightarrow x = \dfrac{1}{2}\ln\left(\dfrac{1+u}{1-u}\right)$	M1 (AO 3.1a) A1 (AO 2.1) M1 (AO 3.1a) A1 (AO 1.1) [4]	Use of exponentials Attempt to find e^{2x}	
		b	$4\tanh^2 x + \tanh x - 3 = 0$ $\Rightarrow (4u-3)(u+1) = 0 \Rightarrow u = \dfrac{3}{4}, (-1)$ $\Rightarrow x = \dfrac{1}{2}\ln\left(\dfrac{1+\frac{3}{4}}{1-\frac{3}{4}}\right) = \dfrac{1}{2}\ln 7$ oe i.e. $a = \dfrac{1}{2}, b = 7$ oe	M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [4]	Solve quadratic Substitute into the standard form	
		c	E.g. Because $-1 < \tanh x < 1$	B1 (AO 2.4) [1]	e.g. One root has been rejected as outside range.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
12		a	(i) $\frac{d^2\theta}{dt^2} = -\left(\frac{5}{2}\right)^2 \theta$ $\theta = A \cos \omega t + B \sin \omega t$ or $R \cos (\omega t + \phi)$ $\theta = A\cos\frac{5}{2}t + B\sin\frac{5}{2}t$ or $R\cos\left(\frac{5}{2}t+\phi\right)$	M1 (AO 1.1) A1 (AO 1.1) [2]		If M0 then SC1 for $\theta = A\sin\frac{5}{2}t$ or $\theta = A\sin\frac{5}{2}t$	
			(ii) The model predicts infinite oscillations of the same amplitude; in practice the amplitude must decrease over time.	E1 (AO 3.5b) [1]			
		b	(i) AE: $4m^2 + 16m + 25 = 0$	M1 (AO 1.1a)	Writing down the AE correctly or using $\theta = Ae^{mt}$ and substituting into (*) to derive a three term quadratic AE.		
			$-2 \pm \frac{3}{2}i$ $\theta = e^{-2t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$	A1 (AO 1.1) A1ft (AO 1.1) [3]	Their $e^{pt}(A\cos qt + B\sin qt)$ for solution of AE = $p \pm qi$		
			(ii) $t = 0$, $\theta = 0.9 \Rightarrow A = 0.9$ $\frac{d\theta}{dt} = -2e^{-2t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$ $+e^{-2t}\left(-\frac{3}{2}A\sin\frac{3}{2}t + \frac{3}{2}B\cos\frac{3}{2}t\right)$ $t=0,\ \frac{d\theta}{dt} = 0 \Rightarrow -2A + \frac{3}{2}B = 0$	B1 (AO 3.4) M1 (AO 1.1a) M1 (AO	Attempt to differentiate		

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$B = 1.2$ $\theta = e^{-2t}\left(0.9\cos\frac{3}{2}t + 1.2\sin\frac{3}{2}t\right)$ (iii) In the modified model $\theta \to 0$ as $t \to \infty$ oe This is the behaviour we would expect to observe with a real swing door and so the model is an improvement.	3.4) A1 (AO 1.1) [4] B1 (AO 3.5a) B1 (AO 3.5c) [2]	using the product and chain rules (A may be replaced by a number). Substituting $t = 0$ into $\frac{d\theta}{dt}$ to derive an equation in (A and) B ie the amplitude decays etc	
		c	Need $4m^2 + \lambda m + 25 = 0$ to have repeated solutions so $\lambda^2 - 4\times4\times25 = 0$ $\lambda > 0 \Rightarrow \lambda = 20$	M1 (AO 3.5c) A1 (AO 3.3) [2]	Using "$b^2 - 4ac$" = 0 directly Not −20 or ±20	or $\left(2m + \frac{\lambda}{4}\right)^2 + 25 - \frac{\lambda^2}{16} = 0$ and equating part outside brackets to 0
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13			Limits 1 & 3 seen $V=\pi\int_1^3\left((x-3)\sqrt{\ln x}\right)^2\mathrm{d}x=\pi\int_1^3(x-3)^2\ln x\mathrm{d}x$	B1 (AO 3.1a) M1 (AO 3.1a)	Correct substitution into formula (ignore limits) and simplification to integrable (by parts) form	
			$V=\pi\left(\left[\frac{1}{3}(x-3)^3\ln x\right]_1^3-\int_1^3\frac{1}{3}(x-3)^3\frac{1}{x}\mathrm{d}x\right)$	*M1 (AO 1.1a)	Integration by parts with $(x-3)^2$ (may be expanded) being integrated. May come implicitly from previously expanded form	ie from $\int_1^3 x^2\ln x-6x\ln x+9\ln x\mathrm{d}x$ integrated by parts term by term
			$\frac{1}{x}(x-3)^3=x^2-9x+27-\frac{27}{x}$ soi	A1 (AO 1.1)		
			$V=\frac{\pi}{3}\left(\left[(x-3)^3\ln x-\frac{x^3}{3}+\frac{9x^2}{2}+27x-27\ln x\right]_1^3\right)$	A1 (AO 1.1)		
			$\frac{\pi}{3}((3-3)^3\ln 3-\frac{3^3}{3}+\frac{9\times3^2}{2}-27\times3+27\ln 3$ $-(1-3)^3\ln 1+\frac{1^3}{3}-\frac{9\times1^2}{2}+27\times1-27\ln 1)$	dep (AO 1.1) *M1	Completing the integral. NB $\left[(x-3)^3\ln x\right]_1^3=0$ so may be omitted provided it is seen earlier	
			So volume of S is $\frac{\pi}{9}(81\ln 3-80)$ oe	A1 (AO 3.2a) [7]	Correctly dealing with limits	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	$2\sinh x.\cosh x = 2\left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^x + e^{-x}}{2}\right)$ $= \frac{1}{2}\left(e^{2x} - e^{-2x} - 1 + 1\right) = \frac{1}{2}\left(e^{2x} - e^{-2x}\right) = \sinh 2x$	M1 (AO2.1) A1 (AO2.1) [2]	Use of correct exponential form for sinh or cosh AG so evidence of cancellation or difference between squares required	Proof must be properly completed for A1; ie there must be complete reasoning from LHS = … to … = RHS	
		b	$y = a \cosh x - \cosh 2x$ $\Rightarrow \frac{dy}{dx} = a\sinh x - 2\sinh 2x$ $= a \sinh 0 - 2 \sinh 0 = 0$ when $x = 0$	M1 (AO1.1a) A1 (AO1.1) [2]	Diffn and verify = 0 by substitution (which must be seen) oe		
		c	when $x = 0$, $y = a - 1$ so $(0, a - 1)$	B1 (AO1.1) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	$\frac{dy}{dx} = a\sinh x - 4\sinh x\cosh x = 0$ when sinh x = 0 or $\cosh x = \frac{a}{4}$ $\cosh x = \frac{a}{4}$ has two values if $\frac{a}{4} > 1$ and no values if $\frac{a}{4} < 1$ $\frac{a}{4} = 1$ gives the same stationary point as previously identified, so maximum value of a is 4	M1 (AO2.1) A1 (AO2.1) M1 (AO2.2a) A1 (AO2.2a) [4]	For considering the number of possible values of cosh$x$ For full justification	
		e	When a = 6, $\cosh x = \frac{3}{2} \Rightarrow x = \cosh^{-1}\frac{3}{2}$ $y = 6\cosh\left(\cosh^{-1}\frac{3}{2}\right) - \cosh\left(2\cosh^{-1}\frac{3}{2}\right) = \frac{11}{2}$ i.e. $\left(\cosh^{-1}\frac{3}{2}, \frac{11}{2}\right)$	M1 (AO3.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	BC. Substituting in x value. Can be implied by correct answer. Accept awrt 5.500	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	13		
15		a	$\sinh x = \frac{1}{2}\left(e^x - e^{-x}\right)$, $\sinh 3x = \frac{1}{2}\left(e^{3x} - e^{-3x}\right)$ $\sinh^3 x = \frac{1}{8}\left(e^x - e^{-x}\right)^3 = \frac{1}{8}\left(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}\right)$ $= \frac{1}{4}\left(\frac{e^{3x} - e^{-3x}}{2} - \frac{3e^x - 3e^{-x}}{2}\right) = \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x$ $\Rightarrow 4\sinh^3 x = \sinh 3x - 3\sinh x$	M1 (AO1.1a) A1 (AO1.1) A1 (AO2.1) [3]	B1 for one error AG	
		b	Substitute $u = \sinh x$ $4\sinh^3 x = \sinh 3x - 3\sinh x$ $\Rightarrow 16\sinh^3 x + 12\sinh x = 4\sinh 3x$ $u = \sinh x \Rightarrow 4\sinh 3x = 3 \Rightarrow$ $\sinh 3x = \frac{3}{4}$ $\Rightarrow 3x = \ln\left(\frac{3}{4} + \sqrt{1 + \frac{9}{16}}\right) = \ln 2$ $x = \frac{1}{3}\ln 2 \Rightarrow u = \sinh\left(\frac{1}{3}\ln 2\right) = \frac{\left(2^{\frac{1}{3}} - 2^{-\frac{1}{3}}\right)}{2}$	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO3.2a) [5]	Use of $e^{\frac{1}{3}\ln 2} = 2^{\frac{1}{3}}$	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance		
16		a	DR 12 cosh x sinh x or −13 sinh x $\frac{dy}{dx} = 12\cosh x \sinh x - 13\sinh x$ sinh x(12 cosh x − 13) = 0 x = 0, y = −13 $x = \ln\left(\frac{13}{12} + \sqrt{\left(\frac{13}{12}\right)^2 - 1}\right)$ $x = \ln\left(\frac{3}{2}\right)$ $y = -\frac{313}{24}$ or $x = \ln\left(\frac{2}{3}\right)$ $y = -\frac{313}{24}$	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) A1 (AO 2.2a) A1 (AO 2.2a) [9]	Either term correct Both correct Setting to zero and attempting to solve Correct numerical use of formula for $\cosh^{-1}$ (could be ±) or $x = -\ln\left(\frac{2}{3}\right)$ ft from 'their' x From the symmetry of the sinh and cosh functions (could be explicitly calculated) or $x = -\ln\left(\frac{3}{2}\right)$	First solution Second solution Third solution	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	DR $\frac{d^2y}{dx^2}=12\cosh^2 x+12\sinh^2 x-13\cosh x$ $x=0 \Rightarrow \frac{d^2y}{dx^2}=-1<0$ so maximum point $x=\pm\ln\frac{3}{2} \Rightarrow \frac{d^2y}{dx^2}=\frac{25}{12}>0$ so minimum points	M1 (AO 2.5) A1 (AO 3.2a) A1 (AO 3.2a) [3]	Correct values only. Must be explicit. Correct values only. Must be explicit. Must be both.		
			Total	12			

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	$\sinh x = \frac{e^x - e^{-x}}{2}$, $\sinh 3x = \frac{e^{3x} - e^{-3x}}{2}$ $\sinh^3 x = \left(\frac{e^x - e^{-x}}{2}\right)^3 = \frac{1}{8}\left(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}\right)$ $\Rightarrow 4\sinh^3 x = \left(\frac{e^{3x} - e^{-3x}}{2}\right) - 3\left(\frac{e^x - e^{-x}}{2}\right)$ $= \sinh 3x - 3\sinh x$ $\Rightarrow \sinh 3x = 4\sinh^3 x + 3\sinh x$	M1 A1 A1 [3]	Using exponentials for sinh$x$ and expanding cubic Correct cubic oe Must include exponential form of sinh3x Examiner's Comments Most candidates knew what was required of them in part (i) though many did not set out their work very well. Some candidates wrote expressions involving exponentials but did not connect them to the hyperbolic functions. Many candidates were unable to expand $(e^x - e^{-x})^3$ directly, instead squaring first and then multiplying out with the third factor. This extra unnecessary algebraic manipulation usually gave rise to more errors.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$4w^3 + 3w - 3 = 0$ $\Rightarrow 4\sinh^3 x + 3\sinh x - 3 = 0$ $\Rightarrow \sinh 3x = 3$ $\Rightarrow 3x = \ln\left(3 \pm \sqrt{10}\right)$ $\Rightarrow x = \frac{1}{3}\ln\left(3 + \sqrt{10}\right)$ $\Rightarrow \left(w = \sinh\left(\frac{1}{3}\ln\left(3+\sqrt{10}\right)\right)\right)$	M1 M1 A1 A1 [4]	Make substitution Use result of (i) Must have plus sign. Ignore subsequent working Examiner's Comments In part (ii) full marks were given to the expression in the correct format for either *x* or *w* and very few failed to gain full marks for this part.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18			$y = 3\cosh x - 2\sinh x$ $\Rightarrow \frac{dy}{dx} = 3\sinh x - 2\cosh x$	M1	Correct diffn and soi set = 0	Alternatively: $y = \frac{1}{2}(e^x + 5e^{-x})$ $\Rightarrow \frac{dy}{dx} = \frac{1}{2}(e^x - 5e^{-x}) = 0$
			$= 0 \Rightarrow \tanh x = \frac{2}{3}$	A1		when $e^x = 5e^{-x}$ $\Rightarrow e^{2x} = 5 \Rightarrow x = \frac{1}{2}\ln 5$
			$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right) = \frac{1}{2}\ln 5$	A1	$= \ln\sqrt{5}$ www	If y is wrong then M0
				[3]	Examiner's Comments This first question was mostly done well. A few candidates thought that $\cosh x$ differentiated to give $-\sinh x$ and some used exponentials but then could not simplify $e^x = 5e^{-x}$. It was acceptable for the formula for $\tanh x$ from the formula book to be used and so the conversion to exponentials was a rather long winded method.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sinh^{-1}(2x)\right)$ $=\frac{1}{\sqrt{x^2+(\frac{1}{2})^2}}$ $=\frac{1}{\sqrt{x^2+(\frac{1}{2})^2}}\times\frac{2}{2}=\frac{2}{\sqrt{4x^2+1}}$	M1(AO 1.1a) E1(2.1) [2]	Use formulae from formulae booklet AG At least one intermediate step must be seen	OR M1 $\sinh y = 2x$ so $\frac{\mathrm{d}y}{\mathrm{d}x}\cosh y = 2$ **E1** $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{2}{\cosh y}=\frac{2}{\sqrt{4x^2+1}}$	
		b	$2 - 2x + x^2 = 1+(x - 1)^2$ $\Rightarrow \int \frac{1}{\sqrt{1+(x-1)^2}}\,\mathrm{d}x = \left[\sinh^{-1}(x-1)\right]+c$	B1(AO 3.1a) M1(AO 1.1) A1(AO1.1) [3]	Complete square soi Use correct form oe		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
20			$2\cosh x \sinh x - 3\cosh x$	*M1(AO1.1)	Allow sign errors only	e.g. $\pm 2\cosh x \sinh x \pm 3\cosh x$
			$2\cosh x \sinh x - 3\cosh x = 0$ $\Rightarrow \cosh x(2\sinh x - 3) = 0$	dep*M1(AO2.1)	Their $f'(x) = 0$ and factorising their expression	
			$\cosh x = 0$ has no roots, because $\cosh x \geq 1$	E1(AO2.4)		
			sinh is a 1–1 function so therefore just one stationary point when $(2\sinh x - 3) = 0$	E1(AO2.4)		
			$x = \sinh^{-1}\frac{3}{2} = \ln\left(\frac{3}{2} + \sqrt{1 + \left(\frac{3}{2}\right)^2}\right)$ and completion to $x = \ln\left(\frac{3 + \sqrt{13}}{2}\right)$	E1(AO2.2a)	AG	
			$\cosh x = \sqrt{1 + \sinh^2 x} = \sqrt{\frac{13}{4}}$	M1(AO2.1)	Attempt to evaluate $\cosh x$ and substitute to find y	Or other complete method to find y
			$y = \cosh^2 x - 3\sinh x = \left(\sqrt{\frac{13}{4}}\right)^2 - 3 \times \frac{3}{2}$	A1(AO1.1)	Correct substitution seen	
			completion to $y = -\frac{5}{4}$ oe	E1(AO1.1) [8]	AG	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21			**DR** $[V=]\pi\int_0^4\left(\frac{8}{\sqrt{16+x^2}}\right)^2 dx$	B1(AO1.1)	Must be seen	
			$64\pi\times\frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right)$	M1(AO1.2)	Using formulae booklet	
			$16\pi\tan^{-1}(1)-16\pi\tan^{-1}(0)$	M1(AO1.1)	Substitution of correct limits must be seen	
			$=4\pi^2$	A1(AO1.1)		
				[4]		
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$(e^x + e^{-x})^3 = e^{3x} + 3e^x + 3e^{-x} + e^{-3x}$	M1	Doing the expansion	
		i	$= (e^{3x} + e^{-3x}) + 3(e^x + e^{-x})$	A1		
		i	$\Rightarrow (2\cosh x^3) = 2\cosh 3x + 6\cosh x$	M1	Relating cosh3x to exponentials correctly	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\Rightarrow 8\cosh^3 x = 2\cosh 3x + 6\cosh x$ $\Rightarrow \cosh 3x = 4\cosh^3 x - 3\cosh x$	A1		Examiner's Comments Candidates were advised to begin by expanding $(e^x + e^{-x})^3$ and the majority did as they were instructed. The vast majority achieved, eventually, the required expansion and it remained to convert this expression into the required identity. There are, of course, many different approaches to achieve this but there are one or two general rules to be borne in mind: • state the result you are trying to prove either at the beginning or the end. • make clear what the terms being used represent rather than simply manipulate a range of expressions • avoid a style of proof which end with 0 = 0 QED There were many correct solutions seen although a proportion of these could scarcely be described as succinct. There were other approaches that could lead to success although those employing cosh(A + B) etc would be well advised to know the correct expression.
		ii	$\Rightarrow \cosh 3x = 4\cosh^3 x - 3\cosh x = 6\cosh x$ $\Rightarrow 4\cosh^3 x = 9\cosh x$	M1	Using result of (i)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow \cosh^2 x = \frac{9}{4}$ since cosh x $\neq 0$	A1	At least one rejection needs to be stated. Or cosh $x \geqslant 1$	
		ii	$\Rightarrow \cosh x = (\pm)\frac{3}{2}$ $\quad \cosh x \neq -\frac{3}{2}$	A1		
		ii	$\Rightarrow x = \pm\ln\left(\frac{3}{2} + \sqrt{\left(\frac{3}{2}\right)^2 - 1}\right)$	A1	A1 for each in exact form	
		ii	$= \pm\ln\left(\frac{3}{2} + \frac{1}{2}\sqrt{5}\right)$ or $\ln\left(\frac{3}{2} \pm \frac{1}{2}\sqrt{5}\right)$	A1	Deduct from 5 marks 1 mark for additional incorrect answers	
		ii	Alternative: $\cosh 3x = 6\cosh x \Rightarrow \frac{1}{2}\left(e^{3x} + e^{-3x}\right) = 3\left(e^{x} + e^{-x}\right)$	M1	Using exponentials	
		ii	$\Rightarrow e^{3x} - 6e^{x} - 6e^{-x} + e^{-3x} = 0$ $\Rightarrow e^{6x} - 6e^{4x} - 6e^{2x} + 1 = 0$ let $y = e^{2x}$	A1	Cubic in factorised form	
		ii	$\Rightarrow y^3 - 6y^2 - 6y + 1 = 0 \Rightarrow (y + 1)(y^2 - 7y + 1) = 0$	A1	Rejection of $y = -1$ must be stated.	
		ii	$\Rightarrow y = -1, \frac{7 \pm \sqrt{45}}{2}$	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$e^{2x} \neq -1$ ex $\Rightarrow^{2x}$ $= \frac{7 \pm \sqrt{45}}{2} \Rightarrow x = \frac{1}{2}\ln\left(\frac{7 \pm \sqrt{45}}{2}\right)$ $= \frac{1}{2}\ln\left(\frac{7 \pm 3\sqrt{5}}{2}\right)$	A1	oe in exact form Deduct from 5 marks 1 mark for additional incorrect answers	Examiner's Comments Most candidates used the result proved in part (i) to form a cubic equation in coshx; factorisation was duly carried out and the three values found. Two of the values for coshx were 0 and −1.5 and it was important that candidates indicated that, since coshx ≥ 1, these answers were not valid. A number of candidates either ignored these values without explaining why or tried to find x-values corresponding to either 0 or −1.5. Those who arrived at the only solution of coshx = 1.5 were often able to convert this to $\ln(\frac{3}{2} \pm \frac{\sqrt{5}}{2})$ although the second root was frequently omitted. It was also possible to express the equation in exponentials but this route proved much more complex and those who tried this approach were often unable to form and solve the resulting cubic in e^{2x}. Many of the weakest candidates expressed the equation in exponentials but were unable to make further progress.
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23			$\tanh^{-1} x = y \Rightarrow x = \tanh y = \frac{e^{2y}-1}{e^{2y}+1}$	M1	Oe	A muddle of x and y unless recovered is M0.
			$(e^{2y}+1)x = e^{2y}-1$			
			$e^{2y}(1-x) = (1+x)$			
			$\Rightarrow e^{2y} = \frac{1+x}{1-x}$	A1	Correct expression for e^{2y} oe	
			$2y = \ln\left(\frac{1+x}{1-x}\right)$			
			$\left(y = \tanh^{-1} x\right) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$	A1	ag Examiner's Comments This standard question was usually well done, but often variables became muddled and candidates got lost.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$y=2\sinh x+3\cosh x \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=2\cosh x+3\sinh x$	M1	Diffn and setting = 0	$y=\frac{2}{2}\left(e^x-e^{-x}\right)+\frac{3}{2}\left(e^x+e^{-x}\right)=\frac{1}{2}\left(5e^x+e^{-x}\right)$
		i	$=0$ when $2\cosh x=-3\sinh x \Rightarrow \tanh x=-\frac{2}{3}$	A1	Correct value for sinh$x$, cosh$x$ or tanh$x$	$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}\left(5e^x-e^{-x}\right)$
		i	$x=\tanh^{-1}\left(-\frac{2}{3}\right)=\frac{1}{2}\ln\left(\frac{1-\frac{2}{3}}{1+\frac{2}{3}}\right)=\frac{1}{2}\ln\left(\frac{1}{5}\right)=-\frac{1}{2}\ln 5$	A1	some numerical justification must be seen ag	$=0$ when $5e^x=e^{-x}\Rightarrow e^{2x}=\frac{1}{5}\Rightarrow x=-\frac{1}{2}\ln 5$ Correct exponential form, diffn, set = 0 A1 correct e^{2x}
		i	$\sinh x=\frac{-2}{\sqrt{5}},\cosh x=\frac{3}{\sqrt{5}}\Rightarrow y=\frac{-4}{\sqrt{5}}+\frac{9}{\sqrt{5}}=\sqrt{5}$	B1	Exact answer only	A1 answer
					Examiner's Comments A majority of candidates differentiated correctly and then worked through to the answer in exponentials or by finding a value for tanh$x$ and using the standard ln form. A few attempted to obtain a quadratic in sinh$x$ or cosh$x$. Some quadratics were correct but no candidate was able to progress to find x. Far too often the value of y was given as an approximation, and sometime not a very good one.	SC Substitute given value of x into derivative to get 0 is 1/3
		ii	$2\sinh x+3\cosh x=5\Rightarrow 2\frac{e^x-e^{-x}}{2}+3\frac{e^x+e^{-x}}{2}=5$	M1	Find exponential form	Alt:
		ii	$5e^x+e^{-x}=10$			$\Rightarrow \sqrt{5}\cosh(x+\alpha)=5$ where $\alpha=\frac{1}{2}\ln 5$
		ii	$5e^{2x}-10e^x+1=0$	A1	Correct quadratic	$\Rightarrow x=\ln\left(\frac{2+\sqrt{5}}{\sqrt{5}}\right)$ and $-\ln\left(2\sqrt{5}+5\right)$
		ii	$e^x=\frac{10\pm\sqrt{100-20}}{10}=\frac{10\pm\sqrt{80}}{10}$	M1	Solve *their* 3 term quadratic	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$x = \ln\left(1+\frac{2\sqrt{5}}{5}\right)$ and $\ln\left(1-\frac{2\sqrt{5}}{5}\right)$	A1	oe Single ln only Examiner's Comments Once again the method to find a quadratic in sinh$x$ or cosh$x$ proved unfruitful, but the majority found a quadratic in e^x which as straightforward to solve.	Penalise only once
		ii		A1		
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25			$x^2 + 4x + 8 = (x + 2)^2 + 4$	M1 A1	Complete the square in order to use standard form	
			$\int_0^1 \frac{1}{\sqrt{x^2+4x+8}}\,dx = \int_0^1 \frac{1}{\sqrt{(x+2)^2+4}}\,dx$ $= \left[\sinh^{-1}\frac{x+2}{2}\right]_0^1 = \sinh^{-1}\left(\frac{3}{2}\right) - \sinh^{-1}1$ $= \ln\left(\frac{3}{2}+\sqrt{1+\frac{9}{4}}\right) - \ln\left(1+\sqrt{2}\right) = \ln\left(\frac{3}{2}+\sqrt{\frac{13}{4}}\right) - \ln\left(1+\sqrt{2}\right)$ $= \ln\left(\frac{3+\sqrt{13}}{2+2\sqrt{2}}\right)$	M1 A1 M1 A1	Use correct standard form in integration Answer in $\sinh^{-1}$ form Attempt to turn into log form www isw	
			Alternative for last 4 marks $\int_0^1 \frac{1}{\sqrt{(x+2)^2+4}}\,dx = \left[\ln\left((x+2)+\sqrt{(x+2)^2+4}\right)\right]_0^1$ $= \ln\left(3+\sqrt{13}\right) - \ln\left(2+\sqrt{8}\right) = \ln\left(\frac{3+\sqrt{13}}{2+2\sqrt{2}}\right)$	M1 A1 M1 A1	Attempt to use Standard form Limits www isw Examiner's Comments This was another standard question for which most scored full marks. A number of candidates failed to write down the standard form correctly. This is given in the *List of Formulae*. Candidates should be discouraged from relying on memory when it is not necessary!	
			Alternative for last 4 marks $x+2 = 2\tan\theta \Rightarrow I = \left[\ln\left(\sec\theta+\tan\theta\right)\right]_{\pi/4}^{\tan^{-1}3/2}$ $= \ln\left(\frac{3}{2}+\frac{\sqrt{13}}{2}\right) - \ln\left(1+\sqrt{2}\right) = \ln\left(\frac{3+\sqrt{13}}{2+2\sqrt{2}}\right)$	M1 A1 M1 A1	Substitution Indefinite integral Deal with limits www isw	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$x = \cosh y = \frac{e^y + e^{-y}}{2} \Rightarrow e^y + e^{-y} = 2x$ $\Rightarrow e^{2y} - 2xe^y + 1 = 0$	M1	Finding 3 term quadratic in e^y	
		i	$\Rightarrow e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$ $\Rightarrow y = \ln\left(x \pm \sqrt{x^2 - 1}\right)$	A1	Correct solution	Condone ignoring -ve sign at this point.
		i	Reject – sign as principal value taken	B1	Including reason oe Examiner's Comments Most candidates set up a quadratic equation in e^y and solved it to find e^y and hence y. However, only a few candidates convincingly explained why the positive root needed to be taken, often referring in a vague way to the expression needing to be positive. The best explanations were those that talked about the shape of the graph of $y = \cosh^{-1} x$.	Condone interchange of x and y but final ans must be correct
		i	$\Rightarrow y = \ln\left(x + \sqrt{x^2 - 1}\right)$	A1		
		ii			Alt:	
		ii	$y = \ln\left(x + \sqrt{x^2 - 1}\right) \Rightarrow \frac{dy}{dx} = \frac{1}{\left(x + \sqrt{x^2 - 1}\right)} \times \left(1 + \frac{2x}{2\sqrt{x^2 - 1}}\right)$	M1	$x = \cosh y \Rightarrow \frac{dy}{dx} = \frac{1}{\sinh y}$	
		ii	$= \frac{1}{\left(x + \sqrt{x^2 - 1}\right)} \times \frac{\left(x + \sqrt{x^2 - 1}\right)}{\sqrt{x^2 - 1}} = \frac{1}{\sqrt{x^2 - 1}}$	A1	$= \frac{1}{\sqrt{x^2 - 1}}$ Examiner's Comments Was often answered well, using the derivative of the inverse ($x = \cosh y$) tended to create fewer algebraic errors than direct differentiation.	
		iii	$x = \cosh^{-1} 3$	M1	Use of $\cosh^{-1}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \ln\left(3+\sqrt{8}\right)$	A1		
		iii	$= -\ln\left(3+\sqrt{8}\right)$ **oe**	A1	ft, -ve the first answer Examiner's Comments Several candidates lost the second solution.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	$\cosh x = \frac{e^x + e^{-x}}{2}$, $\sinh x = \frac{e^x - e^{-x}}{2}$	B1	Correct formulae	
		i	$\Rightarrow \cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$	M1	Dealing with squaring correctly	Difference of squares can be used
		i	$= \frac{1}{4}\left(e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}\right) = \frac{1}{4}.4 = 1$	A1	www All steps seen Examiner's Comments The vast majority of candidates scored full marks on this part, either from squaring the exponential forms of the hyperbolic functions or by using the difference of squares.	
		ii	$\Rightarrow \cosh^2 x - 1 = 5 \cosh x - 7$			
		ii	$\Rightarrow \cosh^2 x - 5 \cosh x + 6 = 0$	M1	Use (i)	
		ii	$\Rightarrow (\cosh x - 2)(\cosh x - 3) = 0$	M1	Attempt to solve quadratic	E.g. correct formula or expansion of their brackets gives 2 out of 3 terms correct
		ii	$\Rightarrow \cosh x = 2,3$	A1		
		ii	$\Rightarrow x = \cosh^{-1} 2 = \pm\ln\left(2 \pm \sqrt{3}\right)$	A1	Use correct ln formula	Condone lack of ±
		ii	and $x = \cosh^{-1} 3 = \pm\ln\left(3 \pm \sqrt{8}\right)$	A1	Use correct ln formula Examiner's Comments Only a very few ignored the first part and the majority generally found the solution of the quadratic equation and obtained the result for x.	Condone lack of ±
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$\frac{dy}{dx} = \frac{1}{1-\left(\frac{1-x}{3+x}\right)^2} \times \frac{-(3+x)-(1-x)}{(3+x)^2}$	B1	Sight of standard diffn for $\tanh^{-1}x$	
		i		M1	Fn of fn and diffn of quotient	
		i		A1	Soi correct quotient (i.e. correct expression for 2nd part)	
		i	$\Rightarrow \frac{dy}{dx} = \left(\frac{-4}{(3+x)^2-(1-x)^2}\right) = \frac{k}{1+x}$	A1		
		i	$\Rightarrow \frac{dy}{dx} = \frac{-1}{2(1+x)}$	A1	Correct for y'	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2(1+x)^2}$	A1	2nd diffn (NB AG) Examiner's Comments A significant proportion failed to use the chain rule in the original differentiation and consequently made little progress. Many candidates who made this error did not understand that what they were doing was not going to produce the result required and was involving them in very complicated algebra. Other common errors included not writing down the derivative of $\tanh^{-1} x$ correctly, even though it is in the formula book. In order to obtain the correct result for $f''(x)$ it was necessary to find $f'(x)$. There were some who followed an incorrect route but still came up with the correct $f'(x)$, presumably by working backwards from the given $f''(x)$. This, of course, was not creditworthy. An alternative method was to rewrite $f(x)$ in terms of natural logarithms. For those who could cope with the algebra this often produced a shorter and neater solution.	
		ii	When $x = 0, y = \tanh^{-1}\frac{1}{3}$ or $\frac{1}{2}\ln 2$ or $\ln\sqrt{2}$ $\frac{dy}{dx} = -\frac{1}{2}$	B1	For 1st value (needs to be exact)	
		ii	$\frac{d^2y}{dx^2} = \frac{1}{2}$	B1	For both	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow y = \tanh^{-1}\frac{1}{3} + \left(-\frac{1}{2}\right)x + \left(\frac{1}{2}\right)\frac{x^2}{2}$	M1	Use of correct Maclaurin's series	
		ii	$= \tanh^{-1}\frac{1}{3} - \frac{1}{2}x + \frac{x^2}{4}$	A1	Accept 0.347 Examiner's Comments Those candidates who wrote down the general form of Maclaurin's expansion first, rarely omitted the 2 from the denominator of the third term. Several candidates who were unsuccessful in proving the results in part (i) worked backwards from the given result and were able to pick up the marks in this part.	
			Total	10		